

CORE PLATFORM

OPTIONAL ADD-ON

KEY TECHNOLOGIES

1

PRESENTATION LAYER

(User Interface)

React 18 + TypeScript Web Application

Pages • Components • State Management
Wagmi / Viem • Wallet Integration

AI Agent Interface
(Qwen3-8B)

MCP Tool Server
(17 Banking Tools)

React 18, TypeScript,
Wagmi, Viem,
Ethers.js

2

SMART-CONTRACT LAYER

(On-Chain – EVM)

15 Modular Smart Contracts (Solidity)

Core Four-Tier Hierarchy

World Bank Reserve (Tier 1)
↓
National Banks (Tier 2)
↓
Local Banks (Tier 3)

12 Planned Modules

- Loans, SavingsVault, FixedDeposit
- GroupLendingPool, InterBankPool
- CreditPassport, Compliance (ZKP-KYC)
- Fraud Detection, Oracles, Automation
- and others

Solidity ^0.8.x
OpenZeppelin,
EVM (zkEVM)

3

OFF-CHAIN SERVICES LAYER

(Backend & Data Services)

**Express.js
REST API**
(Node.js)

**PostgreSQL
Relational
Database**

**FastAPI
ML Scoring
Service**

**Chainlink
Oracle
Integration**

**Event
Listener**
(Indexer)

Node.js, Express.js,
PostgreSQL, FastAPI,
Redis, WebSockets

4

CHAINLINK INFRASTRUCTURE

(Oracle & Automation)

Chainlink Decentralized Oracle Services

**Chainlink
Functions**
(Off-Chain Compute)

**Chainlink
Automation**
(Scheduled Tasks)

**Chainlink
Price Feeds**
(Market Data)

**Chainlink
Proof of Reserve**
(Reserve Verification)

Chainlink Functions,
Automation, Price
Feeds, Proof of
Reserve